S&P GLOBAL & CRISIL CAMPUS HACKATHON 2026

RiskPulse AI

From a headline to the positions that carry the exposure

Explainable risk intelligence for a synthetic wholesale-banking portfolio

Local FinBERT | Multi-source fusion | Transmission paths

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PROBLEM AND APPROACH

A headline is not a risk decision

THE GAP

01 Reports arrive without the book they affect.

02 A repeated social post can look like a second event.

03 Sentiment does not name the exposed positions.

04 A loss total cannot be challenged without its assumptions.

THE APPROACH

RiskPulse keeps severity, evidence, exposure, and loss as separate numbers.

1 Ingest news and social text.

2 Fuse copies into one event.

3 Trace one path per position.

4 Reprice that book under an explicit scenario.

Synthetic issuers, positions, and shocks. Public or synthetic text only.

SYSTEM DESIGN

One path, from text to a reconciled loss

01
News and social

02
Clean and extract

03
FinBERT sentiment and event class

04
Fuse copies by credibility and recency

05
Map event to issuer, sector, and asset

06
Stress the synthetic book

07
Attribute the loss

Observations to events

24 → 12

Copied reports collapse

Hero evidence

2 sources

1 distinct text · no extra corroboration

Critical effective risk

8.36 / 10

9.78 × recency × 85.5%

Next.js interface, FastAPI services, SQLite evidence. One local process.

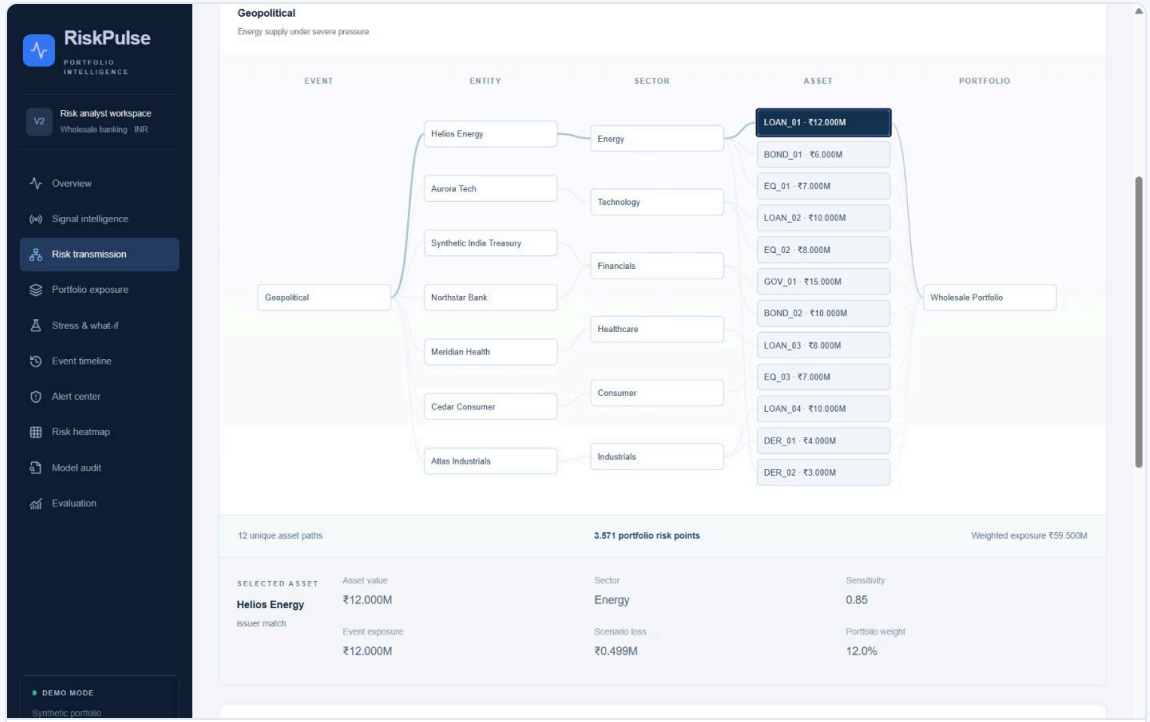
IMPLEMENTATION

Each module answers a risk question

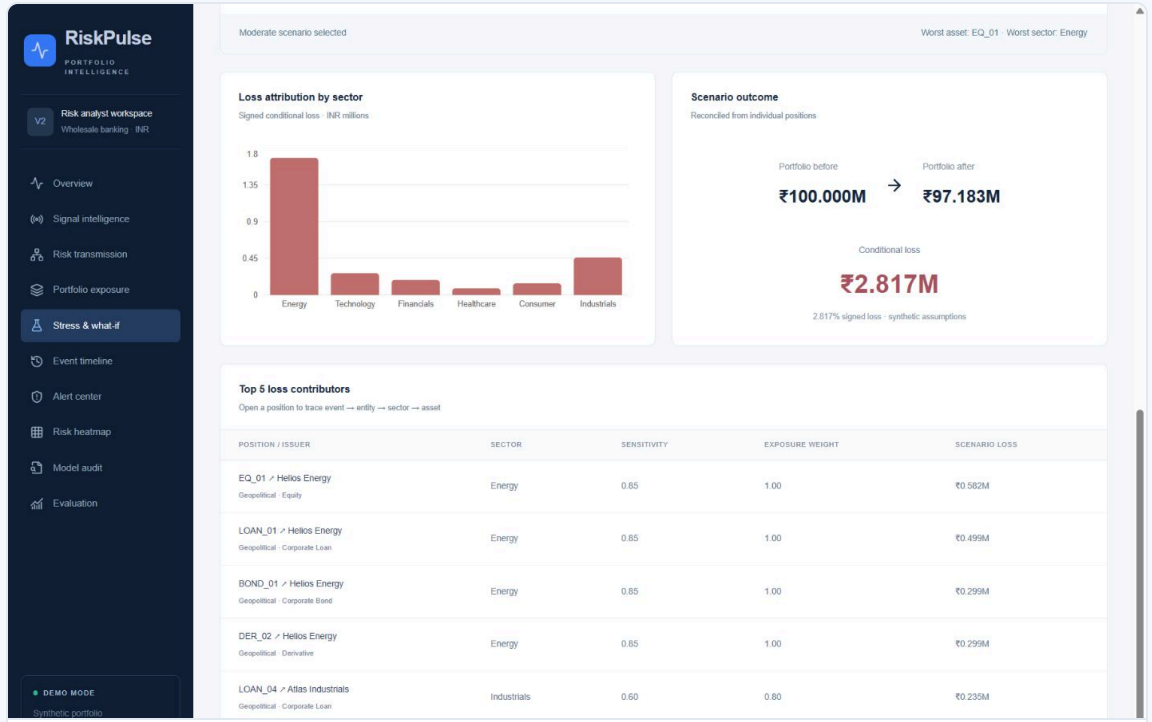
Is this a new event, or the same report again?	**Copy collapse.** Distinct text and publisher can add at most a small evidence bonus.
How severe is it?	**Impact 1–10** from severity, sentiment, relevance, source type, and exposure. Weights are shown.
How strong is the evidence?	**Confidence and recency** adjust priority only.
Which positions are in the path?	**Issuer match, else sector, else systemic spillover.** One path per asset.
What if the shock is stronger?	**Same valuation engine.** Intensity, impact, rates, and equities are explicit inputs.

RESULT

One geopolitical event, traced into the book



RUNNING PROTOTYPE — Risk transmission



RUNNING PROTOTYPE — Stress attribution

Portfolio	Moderate loss	1.5× intensity	Weighted exposure
INR 100M → 97.183M	INR 2.817M	INR 4.226M	INR 59.5M · 12 paths

Conditional synthetic repricing. The moderate case is the original demo result.

DOMAIN IMPACT

The decision is reviewable

PRIORITY

Prioritise events, not headline volume. The copied social post did not raise confidence.

EXPOSURE

See who is exposed before the scenario runs. Direct `INR 42M`, weighted `INR 59.5M`.

CHALLENGE

Challenge the loss. Energy `INR 1.679M`. Largest name `EQ_01` `INR 0.582M`. Audit shows the text, the weights, and the shock.

LOCAL DIAGNOSTIC, NOT A MARKET TEST

`59 tests · event agreement 100% · sentiment agreement 91.67% · 24 synthetic records`

An analyst compares assumptions. The prototype does not forecast prices or capital.

LIMITS AND NEXT STEPS

What this prototype does not claim

NOW

- Small correlated synthetic set
- Dictionary entities
- Heuristic confidence
- Illustrative shocks
- Simplified derivatives, no hedging or netting
- One local process

NEXT, ONLY IF LABELLED AND REVIEWED

- Independent labels
- Source provenance
- Calibrated shocks
- Instrument pricing
- A risk-team backtest

Decision-support prototype. No market-prediction or regulatory-compliance claim.